

TIM 80C, Lecture # 12 (5/7/2026)

Midterm: Created a start-up in the
Hybrid Car Industry

TIM80 C, Lecture # 13 (5/12/2026)

Agenda:

1. Review of Financial Concepts
2. Application of NPV analysis to a start-up.
3. HW # 4 has been posted: please work on it.
4. Read the "Adobe as a start-up" case study on Canvas
5. Study and use the Business plan template on Canvas.

1. Review of financial concepts: done
2. Application of NPV to analyzing cash flows in a start-up

Definition of the problem:

What is the expected profit for a new product development project in a start-up for a time-horizon of " n " years, ($n = 3 - 5$) and an annual discount factor of d % per annum (or year).

Process for constructing a "Base Case" or "Nominal" NPV financial model for Expected Profit.

1. Determine the magnitude (\$) and timing (i.e., when) of the period cash flows
period: quarter or year
2. Determine the net cash flow or profit for each period (usually a quarter)
3. Determine the present value of the net cash flow for each period.
4. Determine the net present value for the project.

Example : Digital Photoprinter at Polaroid.
(on Canvas)

Cash Flows :

Cash Out-flows :

- Product design and Development costs
- Ramp-up costs
- Production costs
- Marketing and support costs

Cash In-flows

- Sales Revenue

Profit = "Cash in-flows" - "cash out-flows"

1 $\equiv$ \$1 Kilodollar = \$1000

Step 2 : Determine the net cash flow or net profit
(for the case study) for each period (in our case, each quarter)

for example : Year 3, Quarter 1

$$\text{Year 3, Qtr 1} \equiv (4 \times 2) + 1 \equiv \underline{\text{Qtr 9}}$$

Period cash flows for Qtr 9 are

$$\text{Marketing cost} = - \$250,000$$

$$\text{Production cost} = - \$2,000,000$$

$$\text{Sales Revenue} = \underline{+ \$4,000,000}$$

$$\left. \begin{array}{l} \text{Net Period (Qtr 9)} \\ \text{cash flow} \end{array} \right\} = + \$1,750,000$$

Step 3 : Present value of each period net cash flow

$$\text{Annual discount factor} = 10\% = d$$

$$\Rightarrow \text{Quarterly discount factor} = \frac{d}{4} = \frac{10}{4} = 2.5\% \equiv$$

$$\text{Present} \equiv \text{Qtr 1}$$

Present value of a future cash flow
is given by

$$P = \frac{F_n}{\left(1 + \frac{d}{4}\right)^{n-1}}$$

where F_n = future cash flow in period
(or quarter) n

P = present value of F_n

d = annual discount factor

For the present example,

$$n = 9; \quad F_n = \$1,750,000; \quad d = 10\% \\ = 0.10$$

$$P (\text{present value of } F_9) = \frac{\$1,750,000}{(1 + 0.025)^{9-1}}$$

$$= \frac{\$1,750,000}{(1.025)^8}$$

$$= \$1,436,306$$

Step 4 :

Add up the present values for each quarter to obtain the net present value for the expected profit, in this case, for

$n = 4$ years, & annual discount factor

$$NPV = (PV)_{Qtr 1} + \dots + (PV)_{Qtr 16}$$

$$= \sum_{i=1}^{n=16} (PV)_{Qtr i}$$

$$= \$8,203K = \$8,203,000$$